

N-S 方程的推导.

一. 前置

① Taylor 展开

$$f(x_0 + \delta x) = f(x_0) + \frac{df}{dx} \Big|_{x_0} \delta x + \frac{d^2 f}{dx^2} \Big|_{x_0} \frac{\delta x^2}{2!} + \dots$$
$$\approx f(x_0) + \frac{df}{dx} \Big|_{x_0} \delta x$$

② 物质导数

对于 $u(x, y, z, t)$

物质导数 $\Downarrow$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz + \frac{\partial u}{\partial t} dt.$$

$$\frac{Du}{Dt} = \frac{du}{dt} = \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w + \frac{\partial u}{\partial t}$$

$$\frac{Du}{Dt} = a.$$

二. 符号.

① $\nabla \cdot$ 散度算子 div (将 = 阶张量 $\rightarrow$ - 阶, - 阶张量 $\rightarrow$ 零阶)

$$\nabla \cdot = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

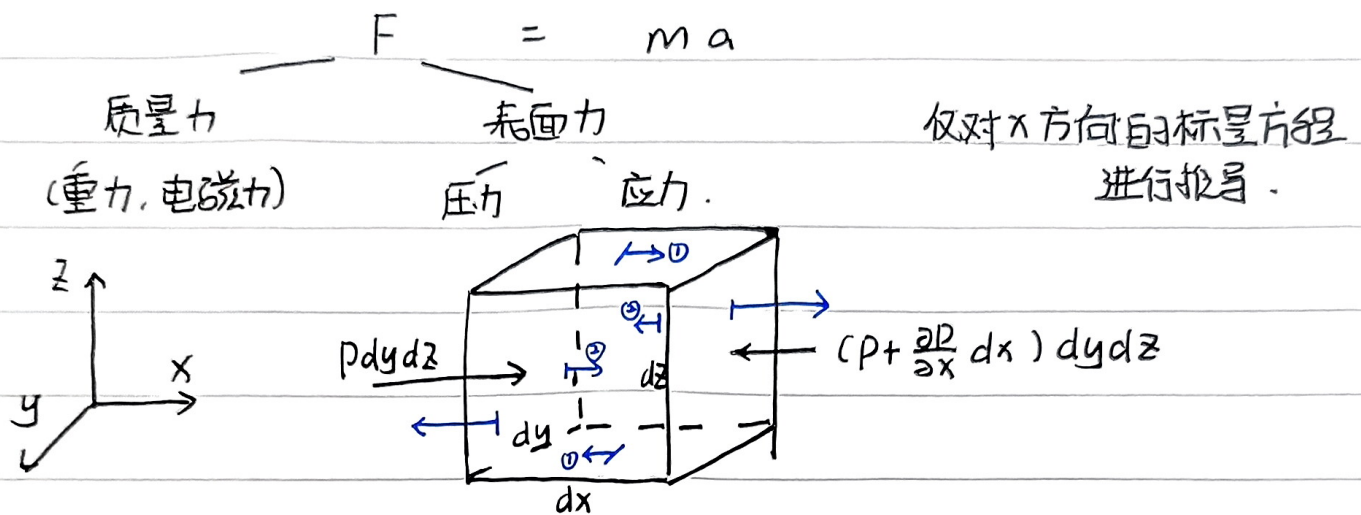
$$\nabla \cdot \underset{\substack{\uparrow \\ \text{-1阶}}}{\vec{V}} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$$

② ∇ 梯度算子 grad (将 - 阶张量 $\rightarrow$ = 阶)

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)^T \quad 3 \times 1$$

$$\nabla P = \left(\frac{\partial P}{\partial x}, \frac{\partial P}{\partial y}, \frac{\partial P}{\partial z} \right)^T \quad 3 \times 1 \quad \text{升阶为矢量.}$$

三. N-S 方程.



① 质量力.

(只和流体微团位置有关)

$$F_x \rho dx dy dz$$

单位质量流体受体积力

② 压力

(指向流体微团中心)

$$p dy dz - (p + \frac{\partial p}{\partial x} dx) dy dz = -\frac{\partial p}{\partial x} dx dy dz$$

运用泰勒展开

③ 应力.

(垂直表面向外)

正应力.

$$\tau_{xx} dy dz$$

$$(\tau_{xx} + \frac{\partial \tau_{xx}}{\partial x} dx) dy dz$$

假设 x, y, z 方向速度 $\uparrow$ 上表面速度 v 大会

把流体微团前拖.

下面 v 小, 后拉

切应力. ①

$$\tau_{zx} dy dx$$

$$(\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} dz) dx dy$$

垂直于受力面轴 力沿该轴

在 z 轴方向变化.

②

$$\tau_{yx} dz dx$$

$$(\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} dy) dx dz$$

$$\therefore F = F_x \rho dx dy dz - \frac{\partial p}{\partial x} dx dy dz + (\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} + \frac{\partial \tau_{yx}}{\partial y}) dx dy dz$$

$$= ma = \rho dx dy dz \frac{Du}{Dt}$$

$$x \text{ 方向: } \rho \frac{Du}{Dt} = F_x \rho - \frac{\partial p}{\partial x} + (\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} + \frac{\partial \tau_{yx}}{\partial y}) dx dy dz$$

$$\vec{U} = (u, v, w)^T \quad \vec{F} = (F_x, F_y, F_z)^T$$

应力表示成张量形式. 不是向量而是矩阵 (2阶张量).

$$\tau = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix}$$

张量形式: $\rho \frac{D\vec{U}}{Dt} = \vec{F}\rho - \nabla P + \nabla \cdot \tau$ N-S方程.

∴ 每一项都应该是一个3x1的向量.

$$\frac{\partial P}{\partial x} \rightarrow \nabla P$$

(0阶张量)

$$\tau \rightarrow \nabla \cdot \tau$$

(2阶张量) (1阶张量)

理论上可求解: $\vec{U}$ 未知, τ 未知.

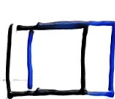
τ 用速度向量表示 $\rightarrow$ 可求解 = 本构方程.

根据牛顿的平板摩擦实验: $\tau = \mu \frac{\partial u}{\partial y}$ 切应力与速度梯度成正比关系.

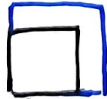
动力粘度 μ $\rightarrow$ $\frac{\text{速度}}{\text{距离}}$ $\frac{\text{相对变化量}}{\text{时间}} \rightarrow$ 相对变化率.

取2阶张量某一项 $\tau_{yx} = \mu \cdot \frac{\partial v_{yx}}{\partial t}$ 相对变化量表示为

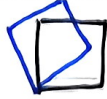
一个流体微团运动可由四种基础运动叠加.



平动



膨胀

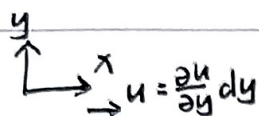


旋转

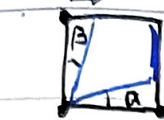


剪切.

只有剪切变形会对切应力产生贡献.



对于剪切



经 dt

 $v=0$ $v = (0 + \frac{\partial v}{\partial x} dx)$ 角度变化量 $r = \beta + \alpha$

$$= \tan \alpha + \tan \beta$$

$$= \frac{\frac{\partial v}{\partial x} dx dt}{dx} + \frac{\frac{\partial u}{\partial y} dy dt}{dy}$$

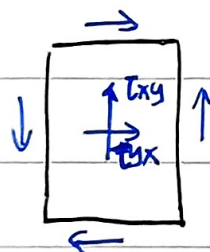
$$= (\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}) dt$$

(对 r 偏导)

$$\therefore \text{角变化率} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

$$\therefore \tau_{yx} = \mu (\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y})$$

$$\tau_{yx} = ? \tau_{xy}$$



力矩相同.

$$\rightarrow (\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} \frac{dy}{2}) dx dz$$

$$\leftarrow (\tau_{yx} - \frac{\partial \tau_{yx}}{\partial y} \frac{dy}{2}) dx dz$$

$$\uparrow (\tau_{xy} + \frac{\partial \tau_{xy}}{\partial x} \frac{dx}{2}) dy dz$$

$$\downarrow (\tau_{xy} - \frac{\partial \tau_{xy}}{\partial x} \frac{dx}{2}) dy dz$$

向内.

力矩方向
垂直纸面向外

$$(\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} \frac{dy}{2}) dx dz \frac{dy}{2} + (\tau_{yx} - \frac{\partial \tau_{yx}}{\partial y} \frac{dy}{2}) dx dz \frac{dy}{2} =$$

$$(\tau_{xy} + \frac{\partial \tau_{xy}}{\partial x} \frac{dx}{2}) dy dz \frac{dx}{2} + (\tau_{xy} - \frac{\partial \tau_{xy}}{\partial x} \frac{dx}{2}) dy dz \frac{dx}{2}$$

$$\therefore \tau_{yx} = \tau_{xy}$$

本
构
方
程.

$$\tau_{yx} = \tau_{xy} = \mu (\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y})$$

$$\tau_{xz} = \tau_{zx} = \mu (\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z})$$

$$\tau_{yz} = \tau_{zy} = \mu (\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z})$$

✓ 可压形式 (加一个膨胀项).

正应力:

$$\begin{cases} \tau_{xx} = \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \right) \mu = 2\mu \frac{\partial u}{\partial x} \left(-\frac{2}{3} \mu \operatorname{div} \cdot \vec{v} \right) \\ \tau_{yy} = 2\mu \frac{\partial v}{\partial y} \quad \leftarrow \text{不可压形式} \\ \tau_{zz} = 2\mu \frac{\partial w}{\partial z} \end{cases}$$

如为可压流体, 会有膨胀变形, 对正应力产生影响.

在研究的大多数实际问题中, 低速流动的情况下,

流体都可以看作不可压流体处理. 散度 $\operatorname{div} \cdot \vec{v} = 0$.
认为

化简.

x方向:

$$\rho \frac{Du}{Dt} = F_x \cdot \rho - \frac{\partial p}{\partial x} + \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} + \frac{\partial \tau_{yx}}{\partial y} \right)$$

✓ 本构方程

$$\mu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} + \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \right].$$

$\nabla \cdot \vec{v} = \operatorname{div} \cdot \vec{v}$ 速度散度.

$\therefore = 0.$

$$\therefore \rho \frac{Du}{Dt} = F_x \rho - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right). \quad \nabla^2 u.$$

向量形式:

$$\rho \frac{D\vec{v}}{Dt} = \vec{F} \rho - \nabla p + \mu \nabla^2 \vec{v} \quad \nabla \cdot (\nabla \vec{v}) = \nabla^2 \vec{v}$$

不可压 NS 方程.